J1A
CardConnector
J1B
CardConnector
J1C
CardConnector
STANDARD CLOCK
SYSTEM CLOCK
Y1
1Mhz
C4
100n
STANDARD_CLK
VISUAL "HEARTBEAT"
DIVIDE CLOCK BY 10*
DEFINITELY OVERKILL BUT FUN
U5A
74HCT390
U5B
74HCT390
U6A
74HCT390
U6B
74HCT390
U7A
74HCT390
U7B
74HCT390
R8
1k
D2
LED
CLOCK SELECTION
AND BUFFERING
SW3
1MS1T1B1M1QES
CLOCK SELECT
R9
1k
R10
1k
C16
100n
C14
10n
U10
555_Alt
CLOCK_SEL
U12A
74HCT151
U13A
74HCT541
R16
33R
PHI2_BO
STANDARD_CLK
DEBUG_CLK
DEBUG CLOCK
ADJUSTABLE "SLOW" CLOCK
0.7Hz - 480Hz
INCREASE RV1 FOR SLOWER INTERVALS
C5
100n
R1
1k
U1
555_Alt
R2
1k
RV1
1M
CLOCK RATE
C1
10n
C2
1u
16V
SINGLE CLOCK
SW1
1MS2T1B1M1QES
SINGLE CLOCK
R3
1k
C10
100n
R5
1k
U8
555_Alt
C9
10n
C11
100n
DEBUG CLOCK SELECT
SW2
1MS1T1B1M1QES
DEBUG SELECT
R6
1k
R7
1k
C13
100n
U9
555_Alt
C12
10n
U3A
U4A
U2B
U3B
R4
1k
D1
LED_R
MANUAL & POWER ON RESET
PWROK
U2A
C15
10n
SW4
1MS2T1B1M1QES
MANUAL RESET
D3
1N4148
R11
R12
C19
100n
≈ 100ms PULSE
C17
1u
C18
10n
R13
R14
R15
4k7
R15 is a 4k7 pull-up resistor located on a bus terminator card.
Q1
2N3904
#RESET
U2C
U2D
U2E
U2F
U3C
U4B
U4C
U3D
U4D
POWER CONDITIONING AND UNUSED GATES
C3
33u
35V
U2G
74HCT14
C6
100n
U3E
74HCT08
C23
100n
U4E
74HCT32
C7
100n
U5C
74HCT390
C8
100n
U6C
74HCT390
C20
100n
U7C
74HCT390
C21
100n
U12B
74HCT151
C22
100n
U13B
74HCT541
C24
100n
CARD MOUNTING BRACKET
MH1
CardBracket

75-1013-01
CLOCK+
IMNOTWOPR
THE HOUSE OF GALAHAD
Sheet: /CLX/
File: CLX.kicad_sch
Title: CLX
Size: A2 | Date: 2026-11-02 | **Rev: A**
KiCad E.D.A. 10.0.5 | Id: 1/2

75-1013-01
CLOCK+
IMNOTWOPR
THE HOUSE OF GALAHAD
Sheet: /SYSTEM/
File: system.kicad_sch

Title: CLX

Size: A2	Date:	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 2/2